Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

```
Ethylene  (C2H4)
Basis set    : cc-pVDZ
Charge       : 0
Multiplicity : 1
Total e-     : 16
n-body order : 1
```

GROUND STATE ENERGIES

```
E0 (Ha)  : -78.05755506
E1 (Ha)  : -77.89240700
Gap (Ha) : 0.16514806
HF  (Ha) : -78.03837969
```

CONVEX-ROOF METHODS

```
Methods run    : gcg
Restarts       : 8

C_LB  : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB  : eigendecomposition bound
```

ACTIVE SPACE CAS(8,5)

```
Frozen core orbs : 4
Frozen e-        : 8
Active e-        : 8
Active orbs      : 5
Spin-orbitals    : 10
Hilbert dim      : 25
```

SPIN PROPERTIES

```
<S2>         : 0.000000
S            : 0.0000
<Sz>         : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)
```

GME CRITERIA

```
C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB
```

DOUBLE OCCUPANCY

```
<D> total  : 4.000000
<D> / site : 0.800000
```

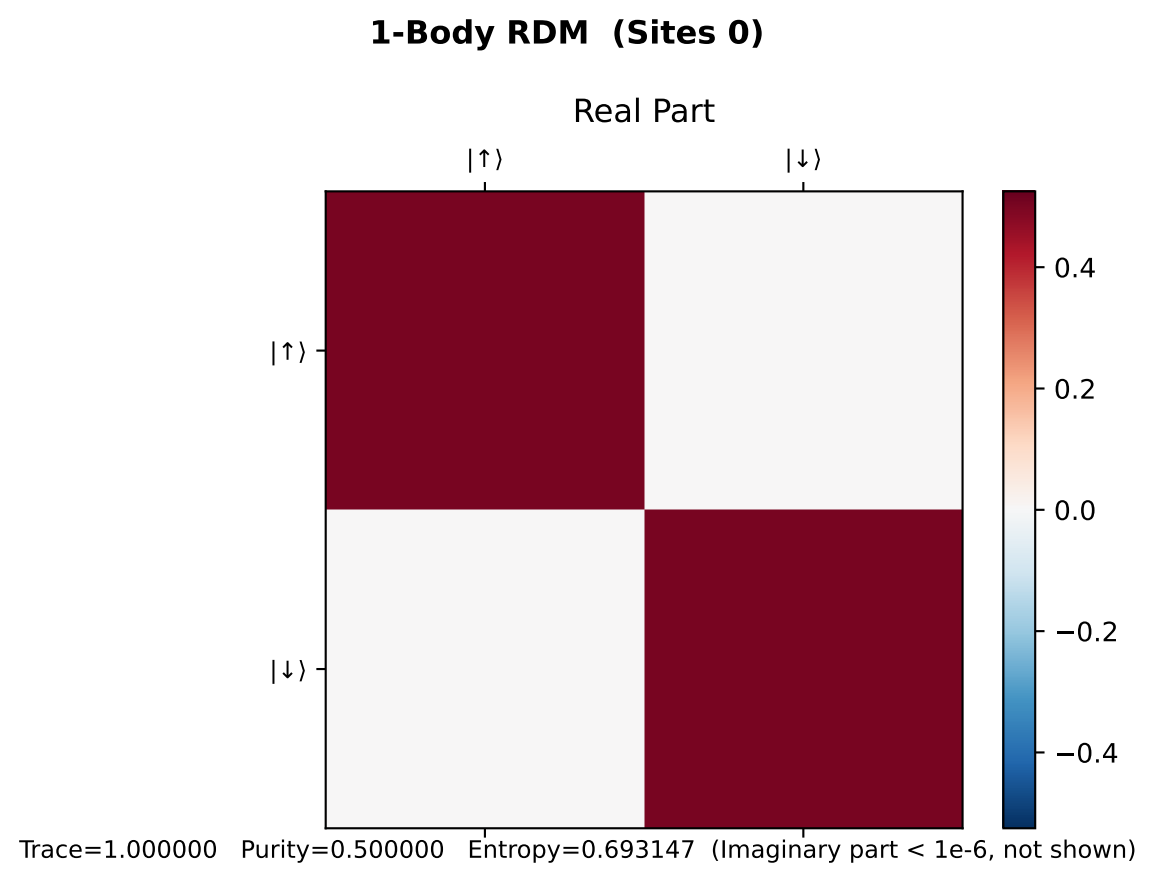
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```
Body order : 1
Clusters   : 5

Date: 2026-07-06  18:15
```

Summary — 1-Body RDMs (Ethylene, CAS(8,5))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0	0.500000	—	—	—	—	—	MIXED
1	0.500000	—	—	—	—	—	MIXED
2	0.500000	—	—	—	—	—	MIXED
3	0.500000	—	—	—	—	—	MIXED
4	0.500000	—	—	—	—	—	MIXED


1-Body RDM (Sites 0)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 0)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

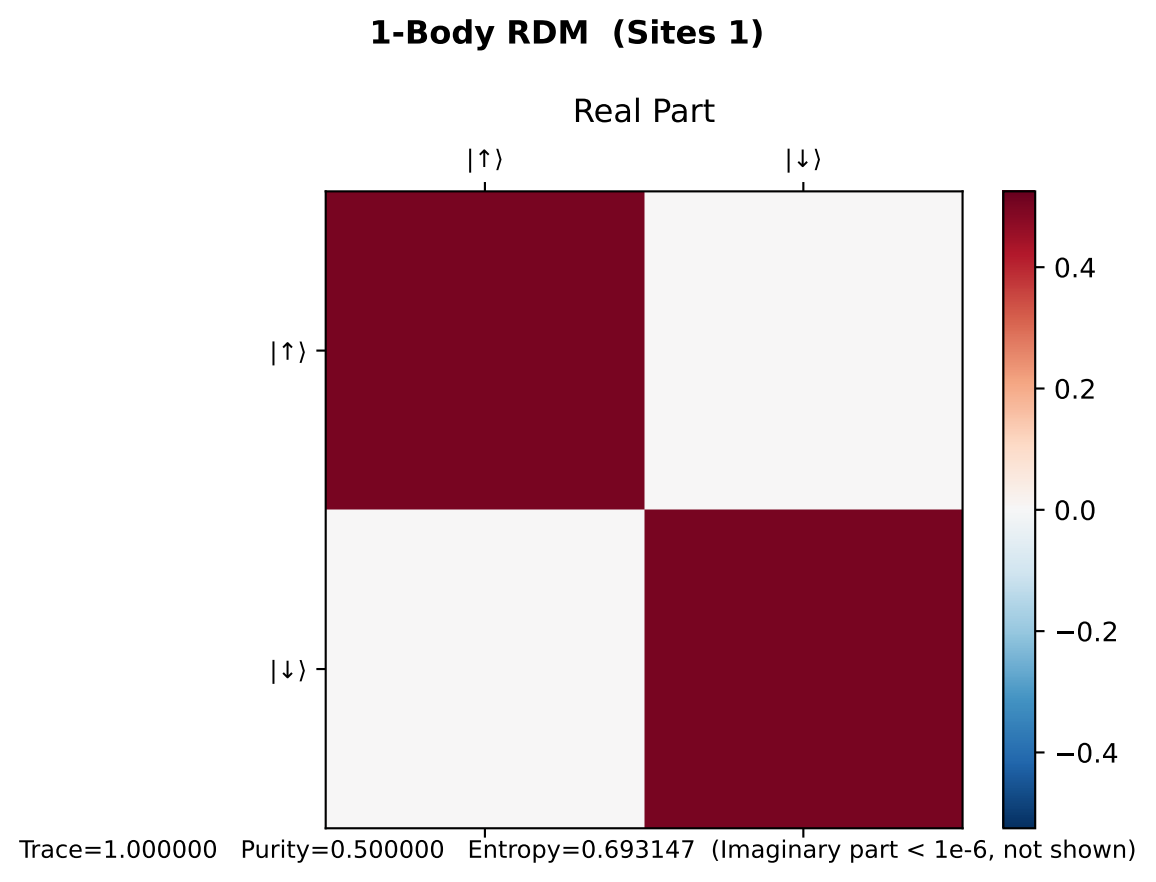
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 1)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 1)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

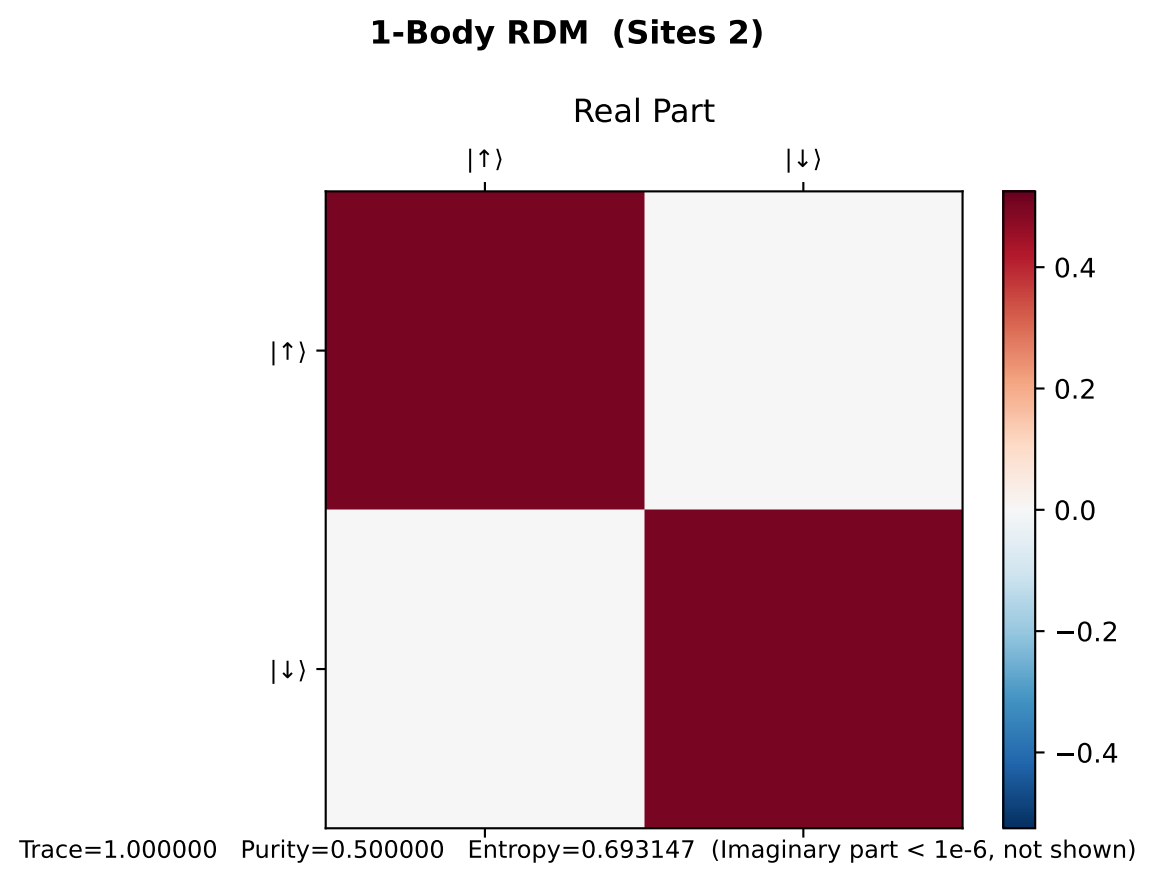
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 2)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

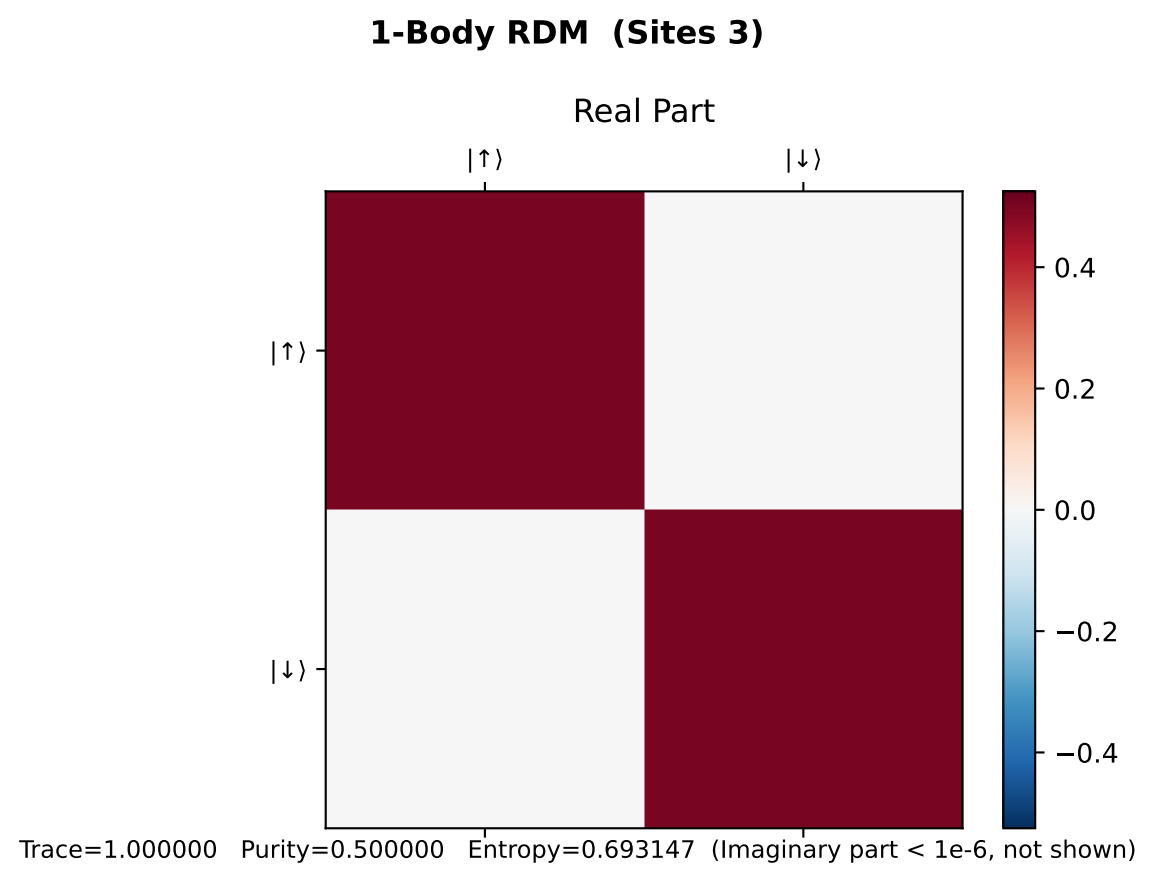
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 3)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

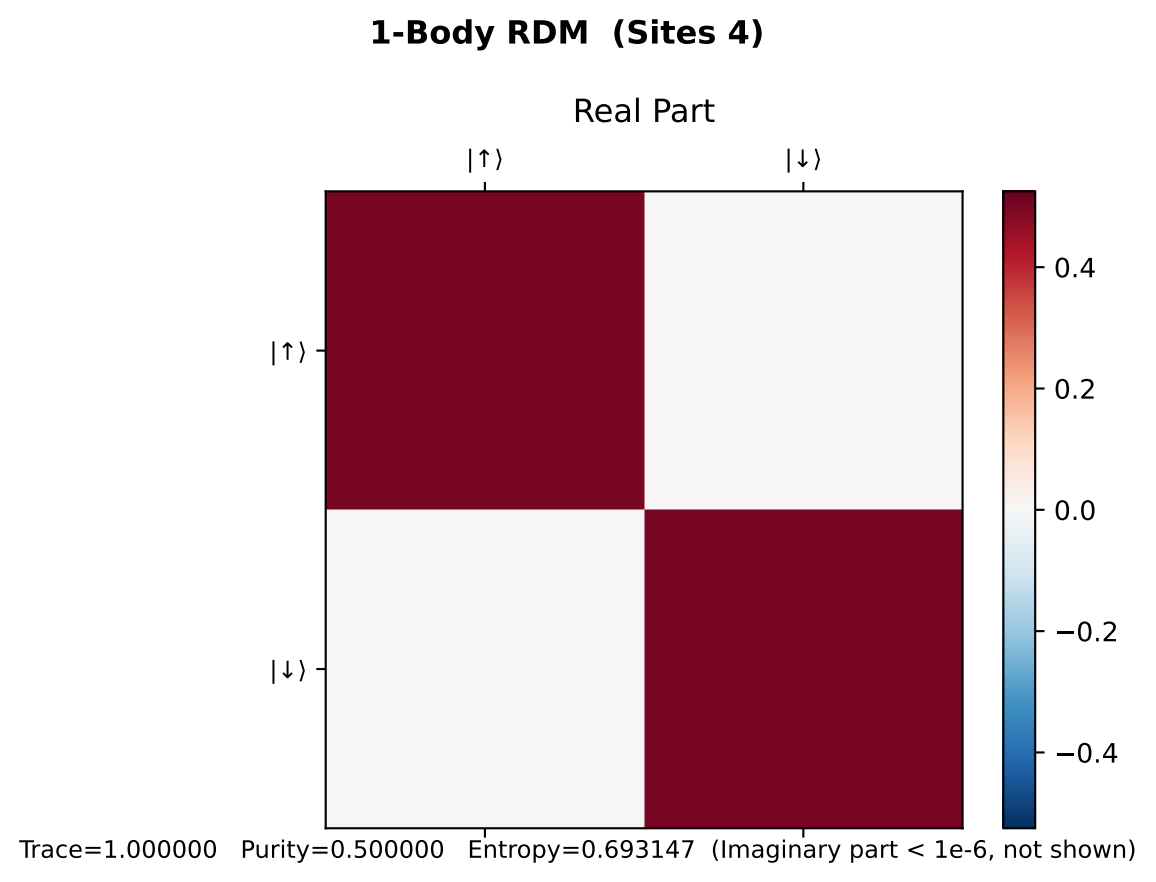
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 4)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.